

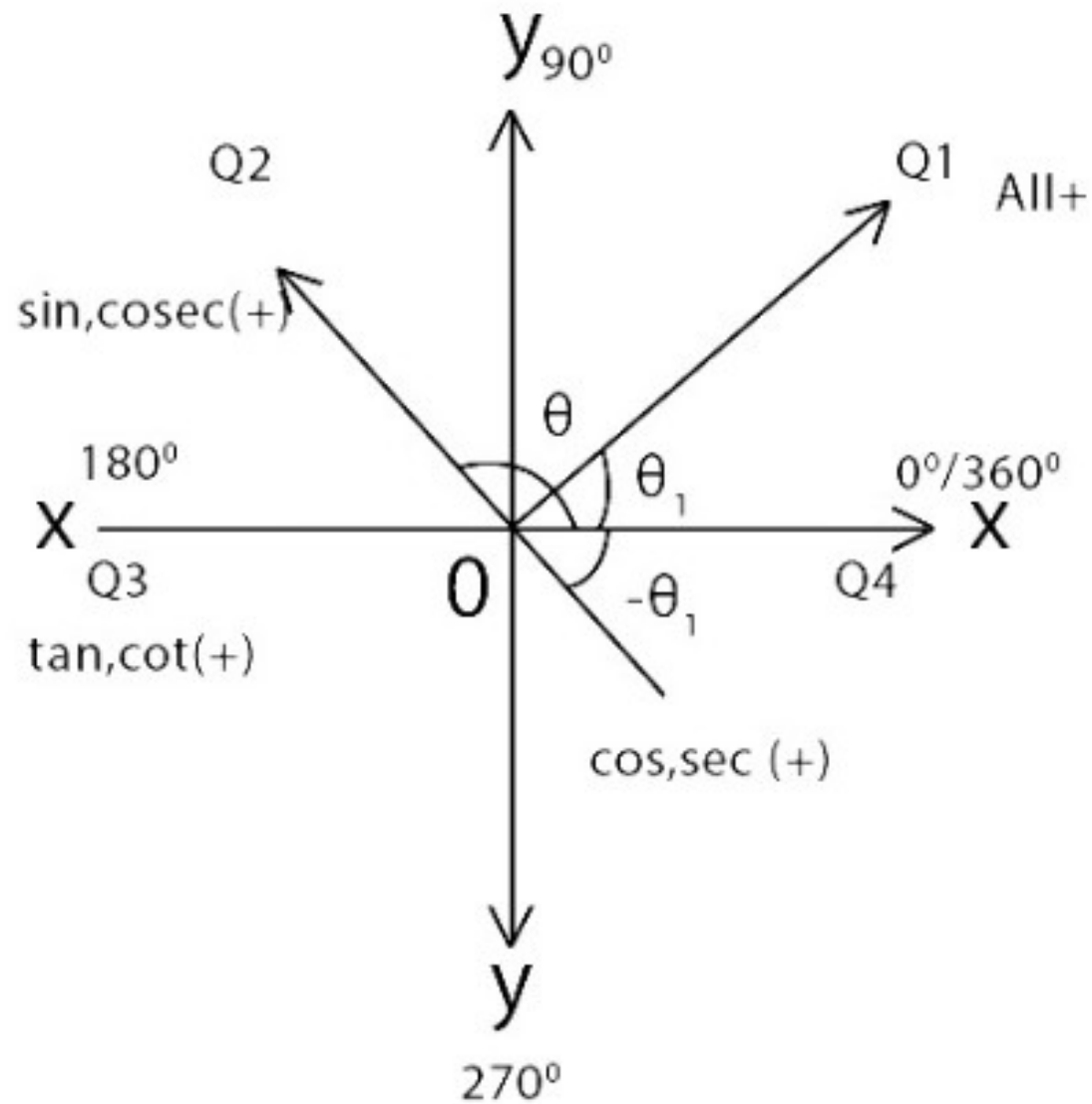
2D Transformation

Translation, Rotation, Scaling,
Shearing, Reflection, Homogeneous

Transformation

Angles (In Degrees)	0°	30°	45°	60°	90°	180°	270°	360°
Angles (In Radians)	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	π	$3\pi/2$	2π
sin	0	1/2	$1/\sqrt{2}$	$\sqrt{3}/2$	1	0	-1	0
cos	1	$\sqrt{3}/2$	$1/\sqrt{2}$	1/2	0	-1	0	1
tan	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞	0	∞	0
cot	∞	$\sqrt{3}$	1	$1/\sqrt{3}$	0	∞	0	∞
cosec	∞	2	$\sqrt{2}$	$2/\sqrt{3}$	1	∞	-1	∞
sec	1	$2/\sqrt{3}$	$\sqrt{2}$	2	∞	-1	∞	1

Transformation



Transformation

- Transformation is the process of relocating, resizing, and changing shape of objects by applying different transformation vectors.
- So, Transformation manipulates the initially created objects.
- Transformation is important as it helps to display the modified object without having to create it from scratch.

Types of Transformation

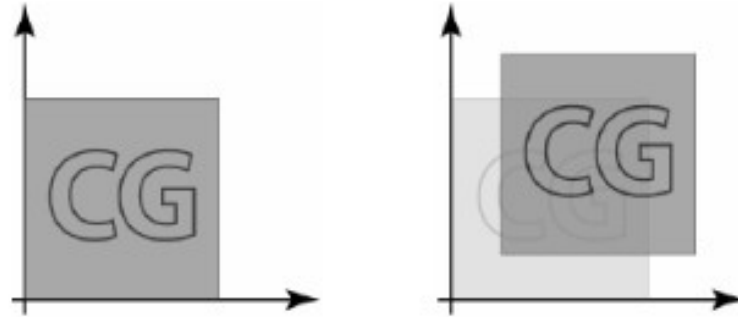
- **Rigid Transformation:** does not change the shape and size of the objects.
- It only changes the location of the object.
- For e.g., Rotation, Translation, Reflection
- **Non-Rigid Transformation:** changes the shape, size, and orientation of the objects.
- For e.g., Shear, Scaling

Types of Transformation

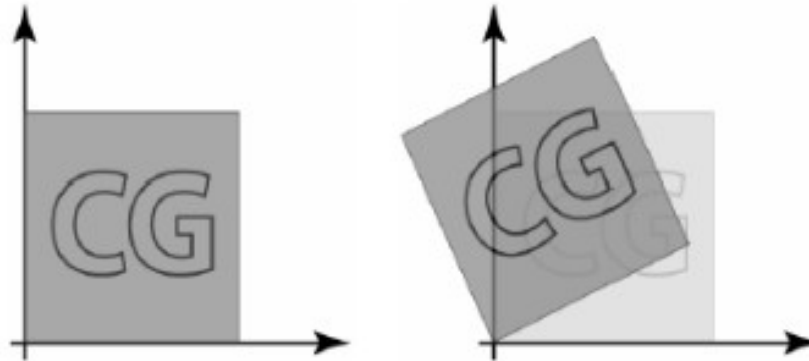
- Another type of transformation are **basic transformation** and **non-basic transformation**
- Application of basic transformation in appropriate way can achieve any type of transformation by applying
- Basic Transformation are
- **Translation**
- **Rotation**
- **Scaling**

Types of Transformation

- Translation

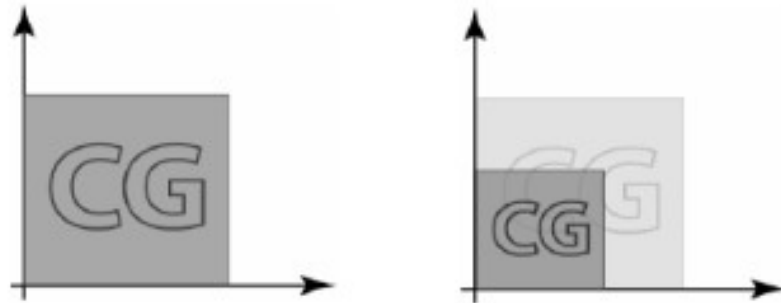


- Rotation

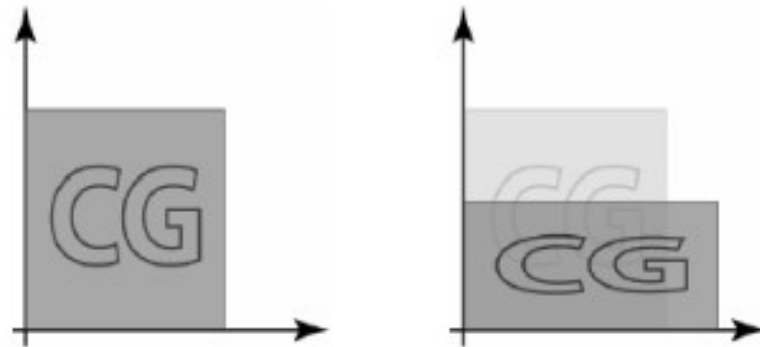


Types of Transformation

- Scaling
- Uniform Scaling

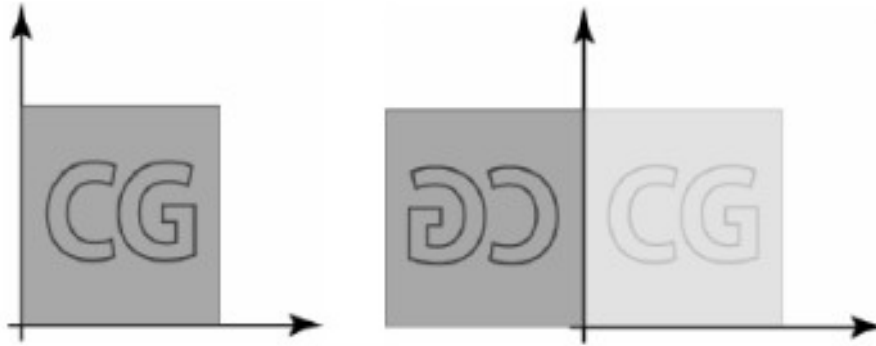


- Un-uniform Scaling

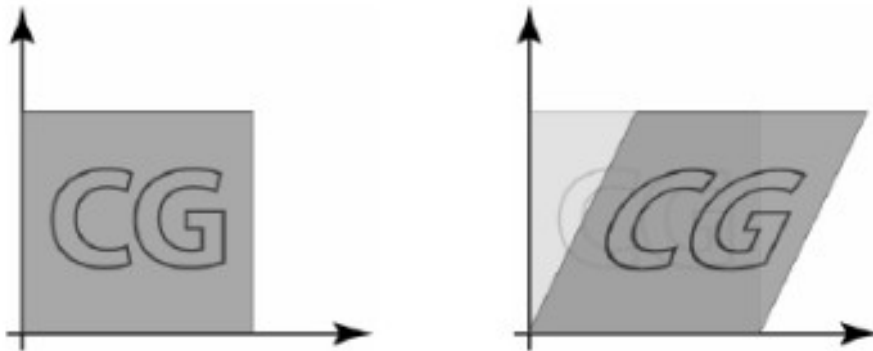


Types of Transformation

- Reflection



- Shear



Translation

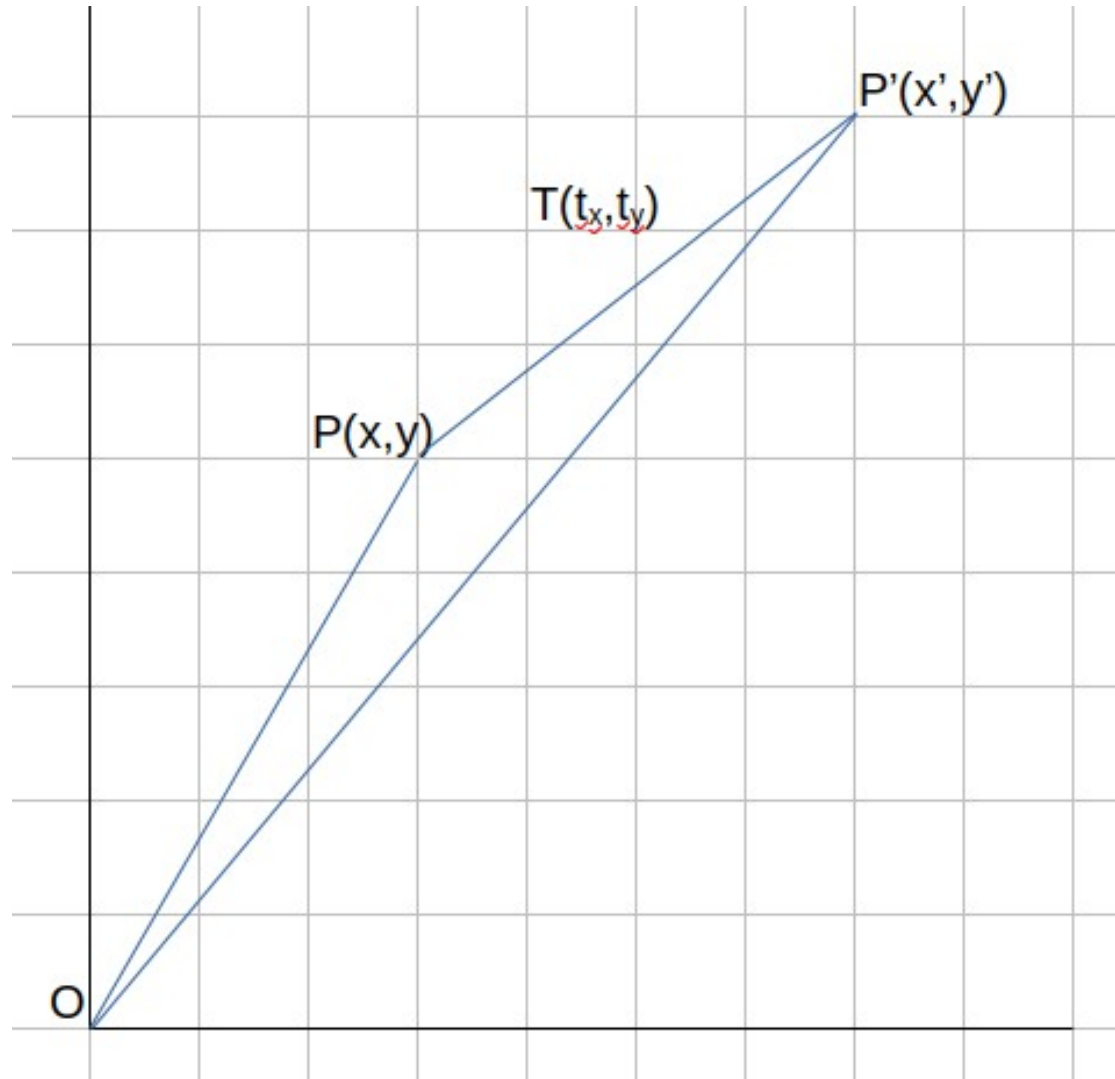
- Translation moves all points in an object along the same straight line path to the new position.
- The path is given by translation vector $T(tx,ty)$.
- Let us translate a point $P(x,y)$ by translation vector $T(tx,ty)$.
- Let the new position be $P'(x',y')$ as shown in the figure.

Translation

- From figure, we can see,
- $OP = (x, y)$
- $PP' = (t_x, t_y)$
- $OP' = (x', y')$
- From triangle law of vectors,
- $OP' = OP + PP'$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \end{bmatrix}$$



Translation

- Translate a point P(4,5) by translation vector T(3,8)
- **Solution:**
- $P' = P + T$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 + 3 \\ 5 + 8 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 7 \\ 13 \end{bmatrix}$$

Translation

- Translate a triangle $A(2,5)$, $B(7,10)$ and $C(10,2)$ by translation vector $T(3,4)$

Translation

$$A' = A + T$$

$$\begin{bmatrix} 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$
$$\begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

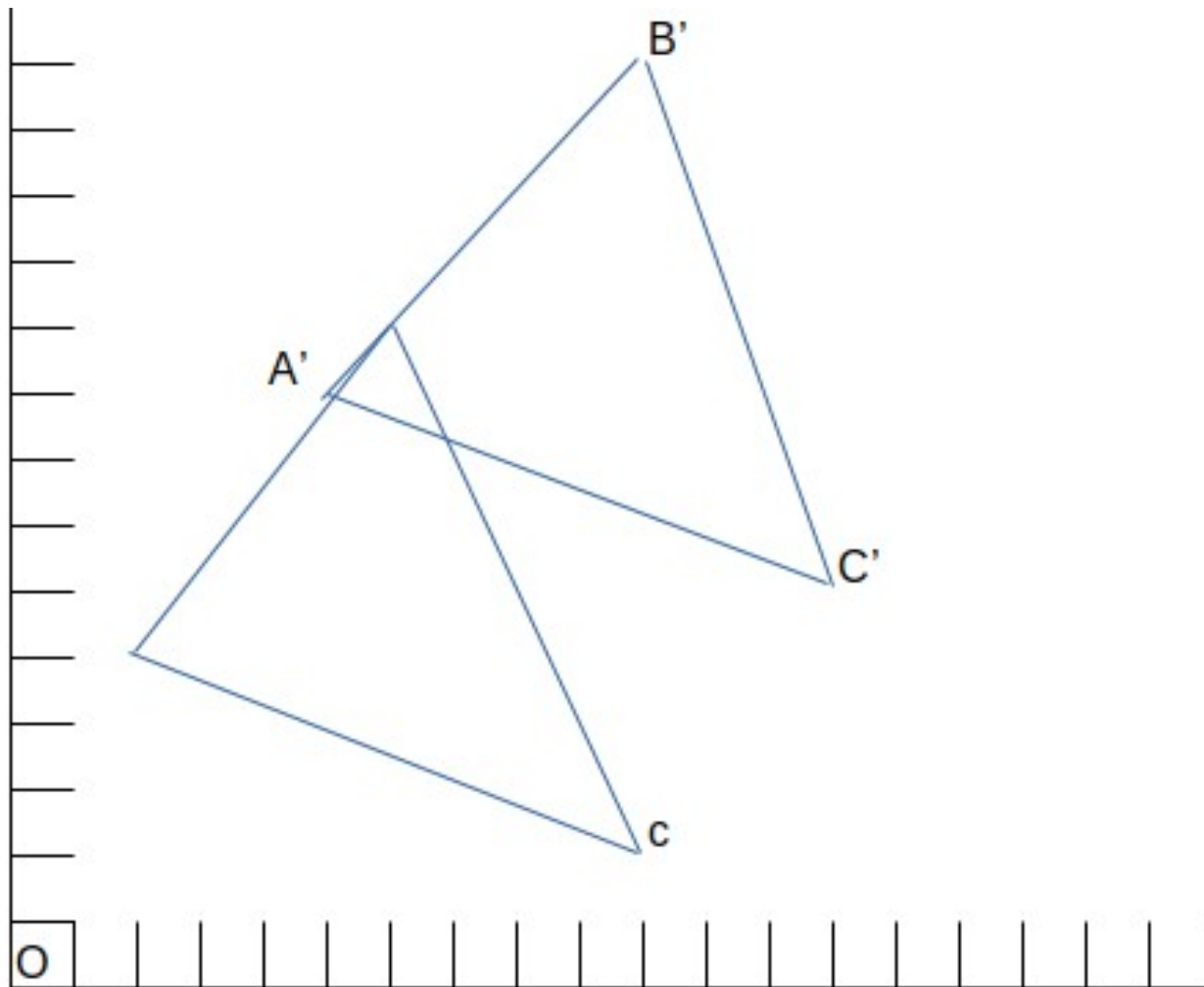
$$B' = B + T$$

$$\begin{bmatrix} 7 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$
$$\begin{bmatrix} 10 \\ 14 \end{bmatrix}$$

$$C' = C + T$$

$$\begin{bmatrix} 10 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$
$$\begin{bmatrix} 13 \\ 6 \end{bmatrix}$$

Translation



Inverse Translation

- Inverse translation of $T(tx,ty)$ is represented by $T^{-1}(tx,ty)$
- $T^{-1}(tx,ty)=T(-tx,-ty)$
- Perform inverse translation by $T(3,4)$ on the triangle $A(5,9)$, $B(13,6)$ and $C(10,14)$.

Rotation

- A rotation repositions all the points in an object along a circular path in the plane centered at the point of rotation (pivot point).
- To rotate a point, we can connect the pivot point and point to rotate with a stick.
- Then rotate one end of stick by desired angle along desired direction by fixing another end at pivot point
- The final position of stick will be final rotated point.

Rotation.

- Let us rotate a point $P(x,y)$ by angle θ anticlockwise with $O(0,0)$ being pivot point.
- Let final point be $P'(x',y')$

$$\text{In } \triangle OAP,$$
$$\cos\phi = \frac{OA}{OP} = \frac{x}{r}$$

$$\text{Or, } x = r\cos\phi$$

Similarly,

$$y = r\sin\phi$$

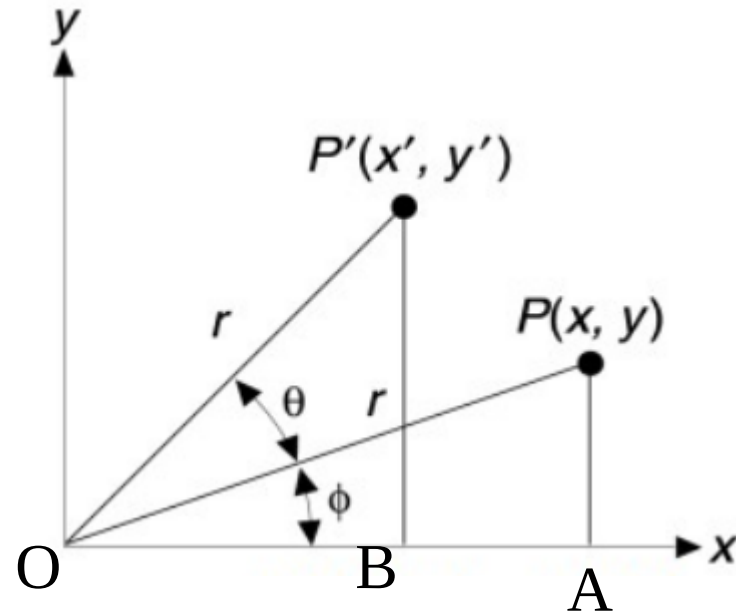
$$\text{In } \triangle OBP',$$

$$\cos(\phi + \theta) = \frac{OB}{OP'} = \frac{x'}{r}$$

$$\text{Or, } x' = r\cos(\phi + \theta)$$

$$\text{Or, } x' = r\cos\phi.\cos\theta - r\sin\phi.\sin\theta$$

$$\text{Or, } x' = x\cos\theta - y\sin\theta$$



Rotation.

- Let us rotate a point $P(x,y)$ by angle θ anticlockwise with $O(0,0)$ being pivot point.
- Let final point be $P'(x',y')$

Similarly,

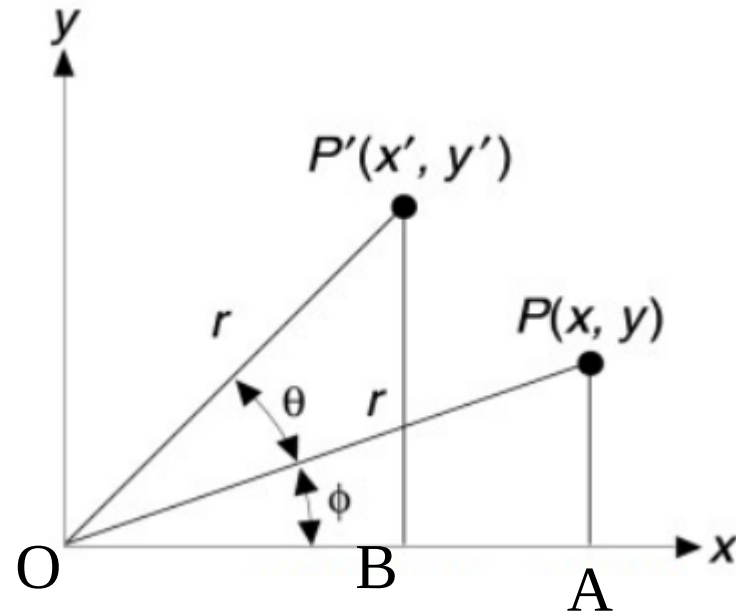
In $\triangle OBP'$,

$$\sin(\phi + \theta) = \frac{P'B}{OP'} = \frac{y'}{r}$$

$$\text{Or, } y' = r \sin(\phi + \theta)$$

$$\text{Or, } y' = r \cos \phi \cdot \sin \theta + r \sin \phi \cdot \cos \theta$$

$$\text{Or, } y' = x \sin \theta + y \cos \theta$$



Rotation.

- In matrix form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- For clock-wise rotation, sign of angle is changed.

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation Example

- A point (4,3) is rotated clockwise direction about origin by 45 degree. Find the resultant point

$$\begin{bmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} \frac{4}{\sqrt{2}} + \frac{3}{\sqrt{2}} \\ -\frac{4}{\sqrt{2}} + \frac{3}{\sqrt{2}} \end{bmatrix}$$

$$\begin{bmatrix} \frac{7}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Inverse Rotation Example

- Perform Inverse rotation

$$\begin{bmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{bmatrix} \begin{bmatrix} \frac{7}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \\
 \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{7}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \\
 \begin{bmatrix} \frac{7}{2} + \frac{1}{2} \\ \frac{7}{2} - \frac{1}{2} \end{bmatrix} \\
 \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Rotation Example

- Rotate a triangle A(2,5), B(7,10) and C(10,2) by 90 about origin anti clockwise

$$\begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 - 5 & 0 - 10 & 0 - 2 \\ 2 + 0 & 7 + 0 & 10 + 0 \end{bmatrix}$$

$$\begin{bmatrix} -5 & -10 & -2 \\ 2 & 7 & 10 \end{bmatrix}$$

Rotation Example

- Inverse Rotation

$$\begin{bmatrix} \cos 90^\circ & \sin 90^\circ \\ -\sin 90^\circ & \cos 90^\circ \end{bmatrix} \begin{bmatrix} -5 & -10 & -2 \\ 2 & 7 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -5 & -10 & -2 \\ 2 & 7 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 + 2 & 0 + 7 & 0 + 10 \\ 5 + 0 & 10 + 0 & 2 + 0 \end{bmatrix}$$

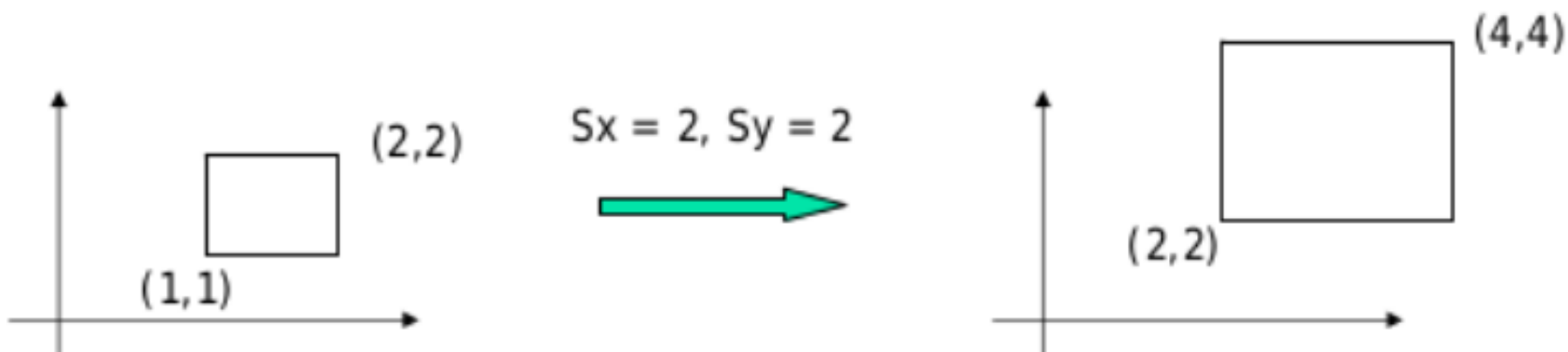
$$\begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

Scaling

- Scaling multiplies the corresponding component by corresponding scaling factor.
- Given a point $P(x,y)$ and Scaling vector $S(s_x, s_y)$, the scaled point $P'(x',y')$ is given by

$$\begin{aligned}x' &= x \cdot S_x \\y' &= y \cdot S_y\end{aligned}$$

$$\Rightarrow \begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} S_x & 0 \\ 0 & S_y \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix}$$



Scaling Example

- Scale a triangle A(2,5), B(7,10) and C(10,2) by scaling vector S(2,3)

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 4 + 0 & 14 + 0 & 20 + 0 \\ 0 + 15 & 0 + 30 & 0 + 6 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 14 & 20 \\ 15 & 30 & 6 \end{bmatrix}$$

Inverse Scaling Example

- Perform inverse scaling

$$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 4 & 14 & 20 \\ 15 & 30 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 2 + 0 & 7 + 0 & 10 + 0 \\ 0 + 5 & 0 + 10 & 0 + 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$



Put it all together

- Translation: $\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} x \\ y \end{vmatrix} + \begin{vmatrix} tx \\ ty \end{vmatrix}$
- Rotation: $\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{vmatrix} * \begin{vmatrix} x \\ y \end{vmatrix}$
- Scaling: $\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} Sx & 0 \\ 0 & Sy \end{vmatrix} * \begin{vmatrix} x \\ y \end{vmatrix}$

Homogeneous Coordinate Space

- Homogeneous coordinate of 2D Cartesian coordinate (x,y) is a 3D vector (x',y',w)
- Where, w is real number,
- $x=x'/w$ and $y=y'/w$
- Generally, we put $w=1$

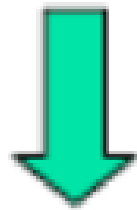
$$\begin{bmatrix} x \\ y \end{bmatrix} \xrightarrow{\text{homogeneous coords}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Advantages of Homogeneous

- All the transformations can be done as matrix multiplication including translation.
- For multiple successive translation, we can pre-compute the combined transformation matrix and apply it to the object.
- This will be efficient.

Translation in Homogeneous

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$$



Use 3 x 1 vector

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation in Homogeneous

- Translate a triangle $A(2,5)$, $B(7,10)$ and $C(10,2)$ by translation vector $T(3,4)$

Translation in Homogeneous

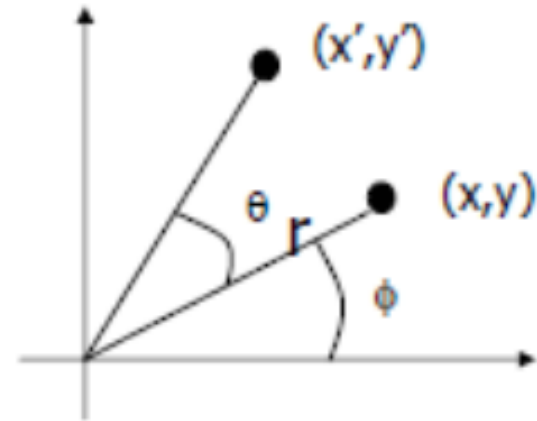
$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 + 0 + 3 & 7 + 0 + 3 & 10 + 0 + 3 \\ 0 + 5 + 4 & 0 + 10 + 4 & 0 + 2 + 4 \\ 0 + 0 + 1 & 0 + 0 + 1 & 0 + 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 10 & 13 \\ 9 & 14 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

Rotation in Homogeneous

$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix}$$



$$\begin{vmatrix} x' \\ y' \\ 1 \end{vmatrix} = \begin{vmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} x \\ y \\ 1 \end{vmatrix}$$

Rotation in Homogeneous

- Rotate a triangle $A(2,5)$, $B(7,10)$ and $C(10,2)$ by 90 about origin anti clockwise using homogeneous coordinate

Rotation in Homogeneous

$$\begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

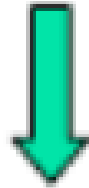
$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 - 5 + 0 & 0 - 10 + 0 & 0 - 2 + 0 \\ 2 + 0 + 0 & 7 + 0 + 0 & 10 + 0 + 0 \\ 0 + 0 + 1 & 0 + 0 + 1 & 0 + 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} -5 & -10 & -2 \\ 2 & 7 & 10 \\ 1 & 1 & 1 \end{bmatrix}$$

Scaling in Homogeneous

$$\begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} Sx & 0 \\ 0 & Sy \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix}$$



$$\begin{vmatrix} x' \\ y' \\ 1 \end{vmatrix} = \begin{vmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{vmatrix} * \begin{vmatrix} x \\ y \\ 1 \end{vmatrix}$$

Scaling in Homogeneous

- Scale a triangle $A(2,5)$, $B(7,10)$ and $C(10,2)$ by $S(2,3)$ using homogeneous coordinate.

Scaling in Homogeneous

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 + 0 + 0 & 14 + 0 + 0 & 20 + 0 + 0 \\ 0 + 15 + 0 & 0 + 30 + 0 & 0 + 6 + 0 \\ 0 + 0 + 1 & 0 + 0 + 1 & 0 + 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 14 & 20 \\ 15 & 30 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

Inverse transformations

Translation :

$$\mathbf{T}^{-1}(t_x, t_y) = \mathbf{T}(-t_x, -t_y)$$

Rotation :

$$\mathbf{R}^{-1}(\theta) = \mathbf{R}(-\theta)$$

Scaling :

$$\mathbf{S}^{-1}(s_x, s_y) = \mathbf{S}\left(\frac{1}{s_x}, \frac{1}{s_y}\right)$$

Multiple Transformation

- First transformation is kept at rightmost
- Last transformation matrix is kept at leftmost
- The object matrix is kept at rightmost.
- Let object matrix be P
- First transformation matrix be T_1
- Second Transformation matrix be T_2
- Then $P' = T_1.P$
- $P'' = T_2.P'$
- $P'' = T_2.T_1.P$
- Or, $P'' = T.P$ where $T = T_2.T_1$

Successive Translation

- Prove that successive translation is additive

$$\mathbf{P}' = \mathbf{T}(t_{1x}, t_{1y})\mathbf{P} \quad \text{first translation}$$

$$\mathbf{P}'' = \mathbf{T}(t_{2x}, t_{2y})\mathbf{P}' \quad \text{second translation}$$

Combined :

$$\begin{aligned} \mathbf{P}'' &= \mathbf{T}(t_{2x}, t_{2y})\mathbf{T}(t_{1x}, t_{1y})\mathbf{P} \\ &= \begin{pmatrix} 1 & 0 & t_{2x} \\ 0 & 1 & t_{2y} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & t_{1x} \\ 0 & 1 & t_{1y} \\ 0 & 0 & 1 \end{pmatrix} \mathbf{P} = \begin{pmatrix} 1 & 0 & t_{1x} + t_{2x} \\ 0 & 1 & t_{1y} + t_{2y} \\ 0 & 0 & 1 \end{pmatrix} \mathbf{P} \\ &= \mathbf{T}(t_{1x} + t_{2x}, t_{1y} + t_{2y})\mathbf{P} \end{aligned}$$

Successive Transformation

Composite translations :

$$\mathbf{T}(t_{2x}, t_{2y})\mathbf{T}(t_{1x}, t_{1y}) = \mathbf{T}(t_{1x} + t_{2x}, t_{1y} + t_{2y})$$

Composite rotations :

$$\mathbf{R}(\theta_2)\mathbf{R}(\theta_1) = \mathbf{R}(\theta_1 + \theta_2)$$

Composite scaling :

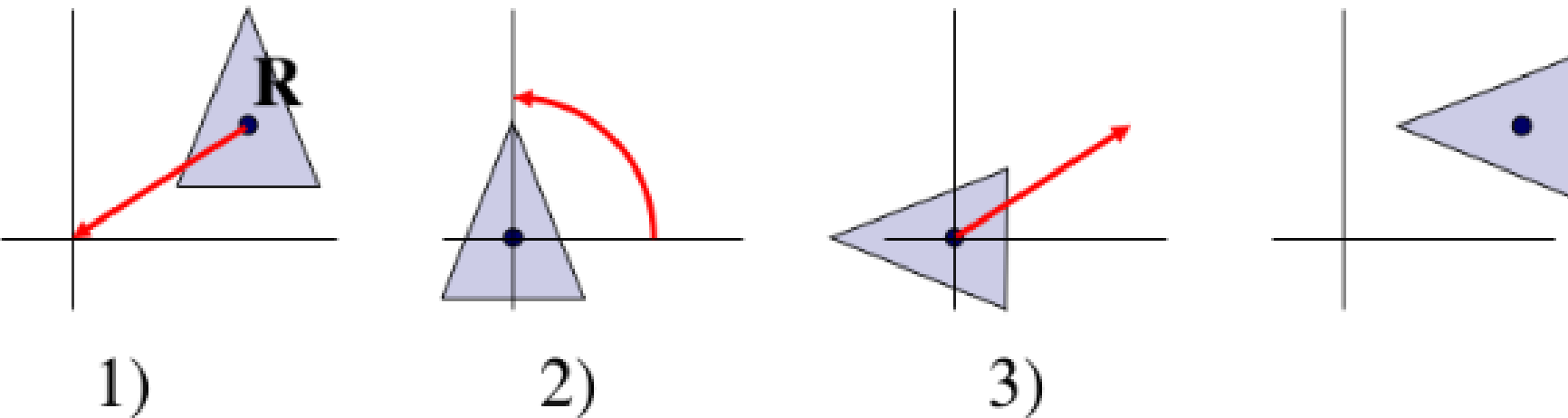
$$\mathbf{S}(s_{2x}, s_{2y})\mathbf{S}(s_{1x}, s_{1y}) = \mathbf{S}(s_{1x}s_{2x}, s_{1y}s_{2y})$$

Rotate about a fixed point

- We have learned about how to rotate an object about an origin.
- Rotating about an origin moves the object from one position to another.
- We want to rotate the object without changing the position.
- For this we have to rotate about the point in the object.

Rotate about a fixed point

- To rotate a point $P(x,y)$ about a fixed point (f_x, f_y) , first translate (f_x, f_y) to origin i.e. $T(-f_x, -f_y)$
- Then rotate the point.
- Perform the inverse translate $T(f_x, f_y)$ to reposition the point at its original position.



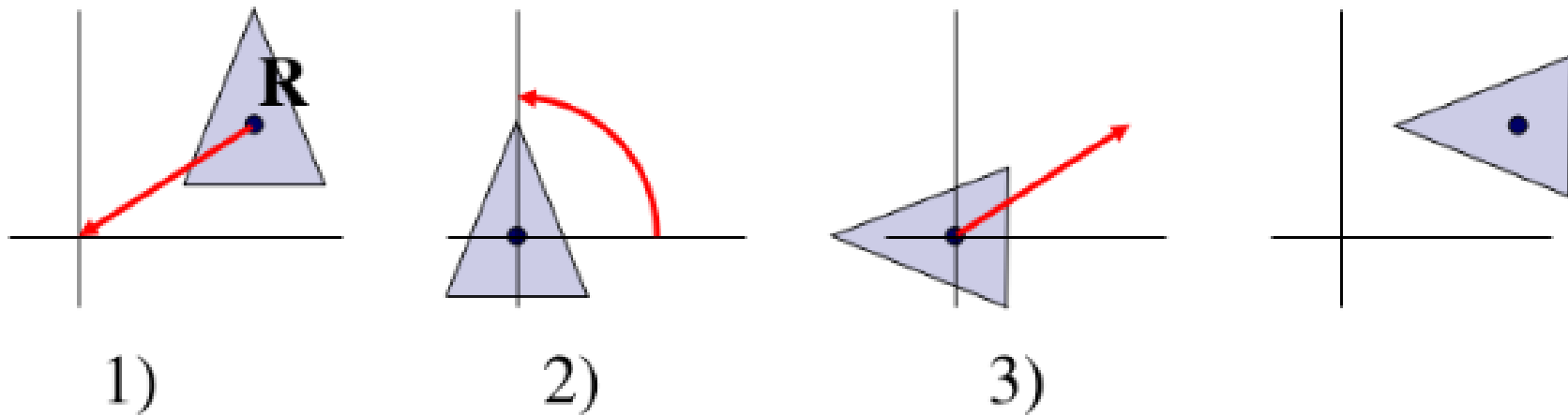
Rotate about a fixed point

Rotate over angle θ around point $\mathbf{R}$:

1) $\mathbf{P}' = \mathbf{T}(-R_x, -R_y)\mathbf{P}$

2) $\mathbf{P}'' = \mathbf{R}(\theta)\mathbf{P}'$

3) $\mathbf{P}''' = \mathbf{T}(R_x, R_y)\mathbf{P}''$



Rotate about a fixed point

$$\begin{bmatrix} 1 & 0 & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -x \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & -x\cos\theta + y\sin\theta + x \\ \sin\theta & \cos\theta & -x\sin\theta - y\cos\theta + y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & x(1 - \cos\theta) + y\sin\theta \\ \sin\theta & \cos\theta & y(1 - \cos\theta) - x\sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

Rotate about a fixed point

- Rotate the triangle $(5, 5)$, $(7, 3)$, $(3, 3)$ about fixed point $(5, 4)$ in counter clockwise (CCW) by 90 degree.

$$\begin{aligned}
& \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \\
& \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \\
& \begin{bmatrix} 0+0+0 & -1+0+0 & 0+0+5 \\ 0+1+0 & 0+0+0 & 0+0+4 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \\
& \begin{bmatrix} 0 & -1 & 5 \\ 1 & 0 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \\
& \begin{bmatrix} 0+0+0 & 0-1+0 & 0+4+5 \\ 1+0+0 & 0+0+0 & -5+0+4 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix} \\
& \begin{bmatrix} 0 & -1 & 9 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned}$$

Rotate about a fixed point

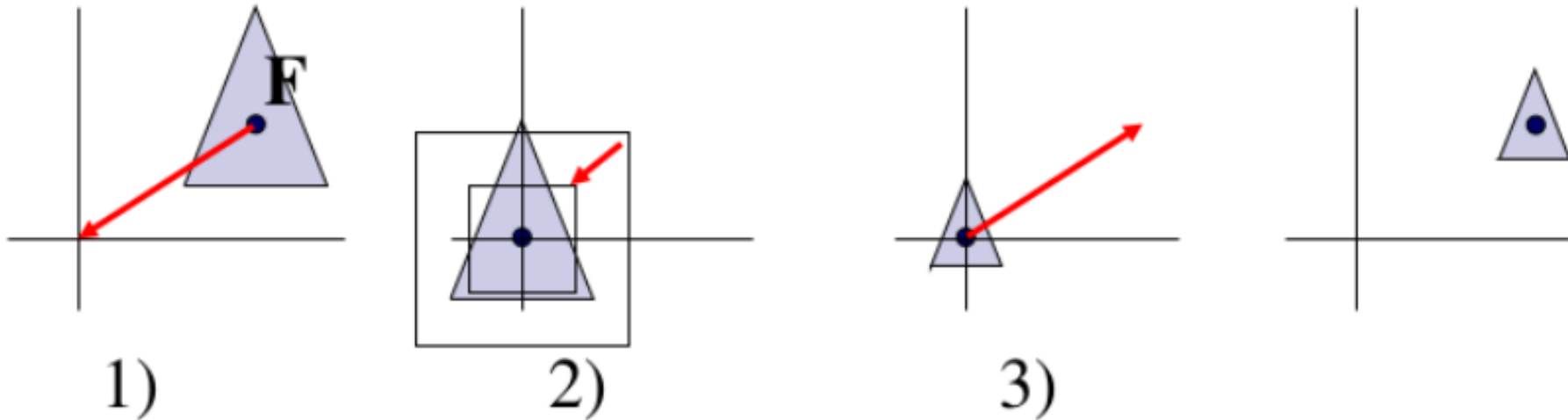
$$\begin{bmatrix} 0 & -1 & 9 \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 7 & 3 \\ 5 & 3 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 - 5 + 9 & 0 - 3 + 9 & 0 - 3 + 9 \\ 5 + 0 - 1 & 7 + 0 - 1 & 3 + 0 - 1 \\ 0 + 0 + 1 & 0 + 0 + 1 & 0 + 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 6 & 6 \\ 4 & 6 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

Scaling about a fixed point

- To scale a point $P(x,y)$ about a fixed point (f_x, f_y) , first translate (f_x, f_y) to origin i.e. $T(-f_x, -f_y)$
- Then scale the point by $S(S_x, S_y)$.
- Perform the inverse translate $T(f_x, f_y)$ to reposition the point at its original position..

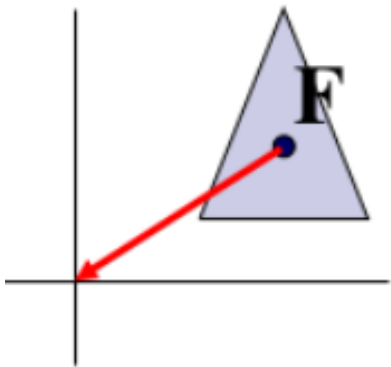


Scaling about a fixed point

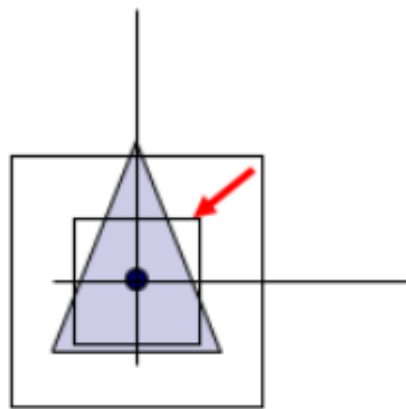
$$1) \mathbf{P}' = \mathbf{T}(-F_x, -F_y) \mathbf{P}$$

$$2) \mathbf{P}'' = \mathbf{S}(s_x, s_y) \mathbf{P}'$$

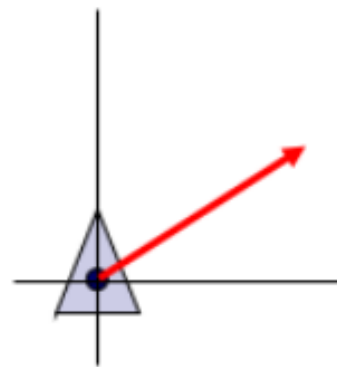
$$3) \mathbf{P}''' = \mathbf{T}(F_x, F_y) \mathbf{P}''$$



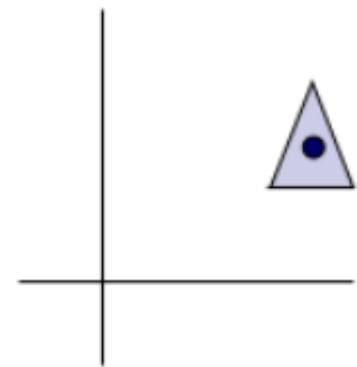
1)



2)



3)



Scaling about a fixed point

- Find the coordinate of a triangle $A(1,3)$, $B(2,5)$ and $C(3,3)$ after scaling twice its original size about fixed point $(2,3)$

Scaling about a fixed point

Matrix for scaling a point (x, y) by (s_x, s_y) about matrix about a fixed point (f_x, f_y) is

$$\begin{bmatrix} s_x & 0 & f_x(1 - s_x) \\ 0 & s_y & f_y(1 - s_y) \\ 0 & 0 & 1 \end{bmatrix}$$

We have, $s_x = 2, s_y = 2, f_x = 2, f_y = 3$

$$\text{So, } \begin{bmatrix} 2 & 0 & 2(1 - 2) \\ 0 & 2 & 3(1 - 2) \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{So, } \begin{bmatrix} 2 & 0 & -2 \\ 0 & 2 & -3 \\ 0 & 0 & 1 \end{bmatrix}$$

Scaling about a fixed point

Now applying this to our triangle.

$$\begin{bmatrix} 2 & 0 & -2 \\ 0 & 2 & -3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 + 0 - 2 & 4 + 0 - 2 & 6 + 0 - 2 \\ 0 + 6 - 3 & 0 + 10 - 3 & 0 + 6 - 3 \\ 0 + 0 + 1 & 0 + 0 + 1 & 0 + 0 + 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 4 \\ 3 & 7 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$